American Computer Science League

2022-2023 ● Contest 1: Short Problems ● Intermediate Division

1. Computer Number Systems

How many 1s are there in the binary representation of the decimal numbers from 8 to 16, inclusive?

A. 23
B. 21
C. 19
D. 17
E. 15

2. Computer Number Systems

Given the following 5 numbers:

$78B_{16}$ ACE_{16} $F4A_{16}$ $2EF_{16}$ BAD_{16}

Which one has the most odd numbers in its octal equivalent?

A. $78B_{16}$
B. ACE_{16}
C. $F4A_{16}$
D. $2EF_{16}$
E. BAD_{16}

3. Recursive Functions

Find $f(25)$ given:

$$f(x) = \begin{cases} 2 & \text{if } x = 1 \\ 4 & \text{if } x = 2 \\ 2 \cdot f(x-1) / f(x-2) & \text{if } x \geq 3 \end{cases}$$

A. 2
B. 4
C. 1
D. 6
E. 3

4. Recursive Functions

Find $f(f(f(9)))$ given:

$$f(x) = \begin{cases} f(x-2) + 2 & \text{if } x \text{ is a multiple of } 4 \text{ and not a multiple of } 3 \\ f(x+1) - 1 & \text{if } x \text{ is a multiple of } 3 \text{ and not a multiple of } 4 \\ x - 3 & \text{otherwise} \end{cases}$$

A. 7
B. 6
C. 4
D. 3
E. 0

5. What Does This Program Do? (Branching)

Which of the following values for a, b, and c will output OBTUSE and SCALENE?

```
input a, b, c
if c >= a + b then
    output "NOT A TRIANGLE"
else
    if a ^ 2 + b ^ 2 == c ^ 2 then
        output "RIGHT"
    else if a ^ 2 + b ^ 2 > c ^ 2 then
        output "ACUTE"
    else
        output "OBTUSE"
    end if
    if a == b and a == c then
        output "EQUILATERAL"
    end if
    if a == b or a == c or b == c then
        output "ISOSCELES"
    else
        output "SCALENE"
    end if
end if
```

A. a=3, b=6, c=9
B. a=4, b=5, c=6
C. a=5, b=6, c=10
D. a=6, b=6, c=6
E. a=6, b=6, c=10

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1. 计算机数字系统

从 8 到 16（**含** 8 和 16）的十进制数字**转换成**二进制**数后，包含**多少个 1？

A. 23
B. 21
C. 19
D. 17
E. 15

2. 计算机数字系统

给定**下面五个数字：**

$78B_{16}$ ACE_{16} $F4A_{16}$ $2EF_{16}$ BAD_{16}

哪一个**表示**为八进制**数字后含有**的奇数最多？

A. $78B_{16}$
B. ACE_{16}
C. $F4A_{16}$
D. $2EF_{16}$
E. BAD_{16}

3. 递归函数

给定下列式子，请计算$f(25)$

$$f(x) = \begin{cases} 2 & \textbf{当} \quad x = 1 \quad \textbf{时} \\ 4 & \textbf{当} \quad x = 2 \quad \textbf{时} \\ 2 \cdot f(x-1)/f(x-2) & \textbf{当} \quad x \geq 3 \quad \textbf{时} \end{cases}$$

A. 2
B. 4
C. 1
D. 6
E. 3

4. 递归函数

给定下列式子， 请计算$f(f(f(9)))$

$$f(x) = \begin{cases} f(x-2)+2 & \text{当 x 是 4 的倍数但不是 3 的倍数时} \\ f(x+1)-1 & \text{当 x 是 3 的倍数但不是 4 的倍数时} \\ x-3 & \text{以上都不满足时} \end{cases}$$

A. 7
B. 6
C. 4
D. 3
E. 0

5. 此程序的作用是什么? （分支）

请问 a、 b 和 c 取**什么值时，会输出** OBTUSE 和 SCALENE？

```
input a, b, c
if c >= a + b then
    output "NOT A TRIANGLE"
else
    if a ^ 2 + b ^ 2 == c ^ 2 then
        output "RIGHT"
    else if a ^ 2 + b ^ 2 > c ^ 2 then
        output "ACUTE"
    else
        output "OBTUSE"
    end if
    if a == b and a == c then
        output "EQUILATERAL"
    end if
    if a == b or a == c or b == c then
        output "ISOSCELES"
    else
        output "SCALENE"
    end if
end if
```

A. a=3, b=6, c=9
B. a=4, b=5, c=6
C. a=5, b=6, c=10
D. a=6, b=6, c=6
E. a=6, b=6, c=10